

DETERMINATION OF ACID-CATALYZED HYDROLYSIS KINETICS IN LOCALLY SOURCED FRUIT EXTRACTS

AIM/ OBJECTIVE

To determine the rate constant of the hydrolysis of Ethyl acetate using an acid as a catalyst.

PRINCIPLE

The hydrolysis of an ester occurs according to the equation



This reaction follows pseudo first order kinetics.

PROCEDURE

100 ml of 0.5 N HCl is taken in a clean dry conical flask. 5 ml of ester is pipetted out into the conical flask and the mixture is immediately withdrawn into another dry conical flask. A stop watch is started simultaneously. The reaction is then arrested by the addition of ice cubes and the mixture is titrated against 0.2 N NaOH using phenolphthalein as indicator. End point is the appearance of permanent pink colour. The volume of NaOH consumed in this titration is taken as V_0 .

5 ml of acid – ester mixture is similarly withdrawn after 10, 20, 30, ..., 60 minutes respectively and titrated against NaOH using phenolphthalein as indicator. The volume of NaOH consumed for each of the above time intervals (t), is taken as V_t .

The contents are transferred into boiling tube with a cap and heated in a water bath for about 15 minutes. 5 ml of this mixture is withdrawn and titrated against NaOH to get V_∞ .

CALCULATION:

The rate constant K is determined using the equation,

$$K = \frac{2.303}{t} \log \frac{(V_{\infty} - V_0)}{(V_{\infty} - V_t)}$$

Rate constant is determined by plotting,

$\log (V_{\infty} - V_t)$ Vs time.

TABULATION:

S.No.	Time Min	Volume of NaOH ml	$(V_{\infty} - V_t)$ ml	$\log (V_{\infty} - V_t)$ ml	$K = \frac{2.303}{t} \log \frac{(V_{\infty} - V_0)}{(V_{\infty} - V_t)}$ min^{-1}
1	0				
2	10				
3	20				
4	30				
5	40				
6	50				
7	60				
8	∞				

RESULT:

The Rate Constant for the hydrolysis of an ester from

1. Calculated value =
2. Graphical value =

MAPPING OF PO & PSO: (For all the COs covered by this experiment)

CO No	PO1	PO2
1	2	1

ASSESSMENT:

Particulars	Max. Marks	Marks awarded
Preparation	10	
Conduct of Experiment	30	
Results & Discussion	30	
Viva - Voce	20	
Report	10	
Total	100	
Evaluator's Signature		

Declaration

The students has completed all the experiments as per the course syllabus on_____

Signature of Faculty